Skills Summary

Reading a Graph Without Being Tricked by It

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the numbers up the side

One step is not always one. Before you read any bar, look at two numbers next to each other up the side.

- $0, 2, 4, 6 \dots$ means one step is worth 2.
- $0, 5, 10, 15 \dots$ means one step is worth 5.
- Then: **steps in the bar** $\times$ **what one step is worth** $=$ **the amount**.

Example: The side numbers go $0, 2, 4, 6, 8$. A bar is 4 steps tall. How much does it show?
Step 1. The bar is being counted in steps, so first find what one step is worth by looking at two side numbers next to each other.
Step 2. From 2 to 4 is one step, so one step is worth 2. Four steps is 4×2, which is **8**.

Try it: The side numbers go $0, 5, 10, 15, 20$. A bar is 3 steps tall. How much does it show?

check: **15**

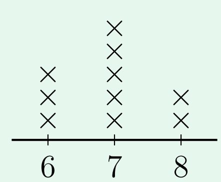
(!) Watch out: Counting the steps is not the answer. It is only half of it. Always turn steps into the thing being counted.

Skill 2 — Counting the marks on a line plot

Two different numbers, two different jobs.

- The number written *under* the line says **what** was measured.
- The marks stacked *above* it say **how many** times that happened. One mark = one thing.

Example: On this line plot, one $\times$ is one bag. How many bags held 7 marbles?



Step 1. The question asks how many *bags*, so the answer comes from counting marks — not from the number printed under the line.
Step 2. Find the 7 on the line and count the marks stacked above it: there are **5**.

Try it: On another line plot, 4 marks sit above the 2, 1 mark sits above the 3, and 6 marks sit above the 4. How many things were counted at 4?

check: **6**

Skill 3 — Naming a shape by counting

Count, then name. Count the straight sides, count the corners, and check whether the sides are all the same length.

- 3 sides: triangle · 4 sides: quadrilateral · 5 sides: pentagon · 6 sides: hexagon.
- A **square** is a quadrilateral with four square corners *and* four sides of the same length. Four sides alone is not enough.

Example: A shape has four square corners and sides of 5, 3, 5, 3. Is it a square?
Step 1. Four sides makes it a quadrilateral, so the question left to settle is whether all four sides are the same length — compare the long side with the short one.
Step 2. $\mathbf{5 > 3}$, so the sides are not all equal. It is a rectangle, not a square.

Try it: A shape has four square corners and sides of 4, 4, 4, 4. Is it a square? Compare a long side with a short side.

check: $\mathbf{4 = 4}$, so yes — it is a square